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A unified model for species fluxes, ion impact energy and heat transfer at the electrode of an atmospheric-pressure arc discharge

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Abstract

1 Introduction

Arc discharges find extensive applications in welding, cutting, and thermal spraying. In recent years, their use has expanded to emerging applications, including plasma pyrolysis, advanced material synthesis, and metallurgical reduction processes [1, 2, 3]. An atmospheric-pressure arc discharge spans a wide range of length scales, differing by several orders of magnitude. The millimeter-to-centimeter-scale quasi-neutral arc column constitutes the bulk of the discharge, where convection, thermal conduction, Joule heating, and radiation interact under conditions that are often close to local thermodynamic equilibrium (LTE). In contrast, the sub-micron to sub-millimeter regions adjacent to the electrodes are characterized by strong nonequilibrium effects. In these regions, the plasma is subjected to the relatively cold electrode surfaces, steep density gradients associated with particle absorption at the electrodes, and strong electric fields arising from charge separation. These fields can accelerate electrons and ions to energies substantially higher than their local thermal energies.

Resolving the full range of length scales throughout the entire discharge is neither computationally practical nor necessary for many applications. Consequently, arc-discharge models commonly treat the quasi-neutral bulk plasma separately from the nonequilibrium electrode regions. In the bulk, the quasineutrality condition, $n_i \approx n_e$, eliminates the need to explicitly solve Poisson's equation, while the near-LTE assumption further reduces model complexity by allowing thermodynamic and transport properties to be determined primarily from the local temperature and pressure. Macroscopic effects such as cathode jets, electrode geometry, convection, and electromagnetic forces can then be incorporated within a continuum formulation based on the Navier–Stokes equations with Joule-heating and Lorentz-force terms. The non-equilibrium electrode regions are generally modeled to account for diffusion, reaction, sheath effects in the discharge.

Quasi-neutral bulk plasma modeling, can be broadly classified into four categories according to the degree of thermodynamic and chemical equilibrium assumed and the treatment of electron and heavy-particle temperatures.

****(i) Full local thermodynamic equilibrium.**** Experimental measurements of species temperatures and electron densities indicate that the arc core is generally close to LTE [4, 5, 6]. In a full LTE model, electrons and heavy particles are characterized by a single temperature T, with Maxwellian velocity distributions, Boltzmann excitation, Saha ionization, and Dalton's law used to determine the plasma state and properties. The governing equations typically consist of mass, momentum, energy, and current continuity [7, 8, 9]. Conventional LTE models does not account for electrode falls and sheaths effects.

****(ii) Two-temperature models.**** Two-temperature (2T) models treat electrons and heavy particles as separate Maxwellian populations at T_e and T_h, coupled through elastic energy exchange. In two-temperature-equilibrium (2T-EQ) models, the chemical composition is additionally determined from a generalized Saha relation for $T_e \neq T_h$. Hsu and Pfender [10] showed that $T_e \approx T_h$ in the high-density arc core, with significant temperature separation emerging in the low-density fringes and near the anode. However, the generalized Saha closure is not unique: different formulations can predict electron densities differing by up to an order of magnitude away

from $T_e/T_h \approx 1$ [11]. Furthermore, inconsistencies in the partitioning of ionization-energy and reactive thermal-conductivity terms between the electron and heavy-particle energy equations have been identified by Freeton et.al [12] in several formulations.

****(iii) Nonequilibrium models.**** Fully nonequilibrium models abandon equilibrium closures such as LTE and generalized Saha equilibrium, allowing the plasma composition and electron and heavy-particle temperatures to evolve independently. These models typically include multicomponent species transport and separate electron and heavy-particle energy equations. Examples include the nonequilibrium arc models of Haidar [13], Gonzalez *et al.* [14], and Amakawa *et al.* [15]. By resolving departures from equilibrium directly, these models can describe the transition from the LTE-like bulk plasma toward the strongly nonequilibrium electrode region. However, the space-charge sheath is generally still treated separately using an appropriate sheath model.

****(iv) LTE with diffusion-driven ionization nonequilibrium.**** Another approach considered in the literature retains $T_e = T_h = T$ while allowing the electron and ion densities to depart from their local Saha-equilibrium values [16, 17, 18]. This departure arises when diffusion competes with the ionization–recombination relaxation processes, particularly near the electrodes. Lowke [16] demonstrated this approach using a full-diffusion calculation that solves the ambipolar electron-continuity equation together with current continuity, including the electron-diffusion contribution. The resulting solution produces a retarding-field potential structure near the electrode that is absent in a conventional LTE calculation. However, because $T_e = T_h$ is retained throughout the domain, this formulation cannot capture the strong electron–heavy-particle temperature separation expected in the low-density electrode-adjacent region.

1.1 Coupling the bulk plasma to the electrode: sheath models

Regardless of the bulk-plasma model employed, the bulk solution must ultimately be coupled to a description of the electrode region. The appropriate coupling strategy depends on the assumptions underlying the bulk formulation and, in particular, on the degree of nonequilibrium that is retained. For fully LTE models (i), the electrode region must account for the breakdown of LTE, including the separation between electron and heavy-particle temperatures, deviations from Saha equilibrium, and the formation of a space-charge sheath. For two-temperature and two-temperature-equilibrium models (ii), the bulk formulation already accounts for electron–heavy-particle temperature nonequilibrium; however, deviations from ionization equilibrium and the formation of a space-charge sheath must still be described. In fully nonequilibrium bulk models (iii), an additional ionization layer or presheath is not necessarily required, since departures from equilibrium are already permitted within the bulk formulation. The principal remaining transition to the electrode is therefore associated with the breakdown of quasineutrality and the formation of the space-charge sheath.

A common approach is to construct the electrode-region model from separate submodels for the ionization layer or presheath and the sheath. The ionization layer is treated as a collisional region and coupled to the sheath, which is treated as collisionless using an analytical model [19, 20, 21, 22]. This approach avoids the direct solution of the stiff Poisson equation across the sheath. However, Benilov *et al.* [23] pointed out that such matching is strictly valid only if the sheath is collision-dominated.

Almeida, Benilov, and Naidis [24] evaluated the number of ion collisions occurring as an ion traverses the cathode space-charge sheath,they pointed out that for atmospheric-pressure argon arc discharge, the ion-collision number is far greater than unity rather than $\ll 1$. Which demonstrate that a physically consistent electrode model for high-pressure arc discharges should account for collisional ion transport within the sheath.

An alternative approach is *unified modeling*, in which a single set of governing equations, including Poisson's equation, is solved across the entire interelectrode region without explicitly separating the bulk plasma from the sheath. This approach provides a self-consistent description of the electrode region but is computationally expensive because the mesh must resolve the extremely small sheath length scales. Consequently, applications to arc discharges have largely been restricted to one-dimensional calculations extending from the electrode into the bulk plasma [24, 25].

Except in the immediate vicinity of the electrodes, an atmospheric-pressure arc is sufficiently close to local thermodynamic equilibrium (LTE) that the bulk plasma can be modeled using a single temperature. Consequently, nonequilibrium effects need not be considered in the bulk, which substantially reduces the complexity of the model.The approach adopted in the present work therefore retains the computationally efficient LTE treatment for the bulk of the arc while explicitly resolving the nonequilibrium physics in the near-electrode region. Rather than

prescribing electrode-boundary quantities or relying on an analytical sheath model, a unified drift-diffusion–Poisson formulation is employed to simultaneously determine the charged-particle distributions, species fluxes, ion impact energies, heat transfer to the electrode, and sheath voltage drop. The same governing drift-diffusion–Poisson system is solved continuously from the quasi-neutral bulk plasma through the near-electrode transition region and into the sheath. The highly disparate spatial scales are resolved using adaptive mesh refinement (AMR), with the computational mesh refined to a fraction of the local Debye length in the sheath region.

An additional is that one need not worry about field reversals. Field reversals and changes in the sign of the potential drop near the anode have been reported in atmospheric-pressure arcs and can depend on several factors, including the attachment mode, current density, and electrode ablation effects [26, 27, 28]. By solving the drift-diffusion–Poisson system self-consistently across the near-electrode region, the sheath potential structure and its polarity emerge from the coupled solution rather than being assumed a priori. This approach enables one to use the results from the computational efficient bulk plasma treatment to inform the boundary conditions for the near-electrode region, while simultaneously capturing the nonequilibrium effects that arise in the sheath and ionization layer. The model and its numerical implementation in the PlasmeX code are described in Section 4.

2 Physical model and scale analysis

2.1 Governing equations

In the electrode region, the plasma interacts with the relatively cold, absorbing electrode surfaces. This interaction introduces strong gradients in the plasma properties, causing the plasma to progressively depart from LTE. We estimate the characteristic length scales associated with these departures by considerin th LTE plasma adjacent to the electrode regions to be present in temperature range $T = 10{,}000$ K to $T = 20{,}000$ K. which are typical of the anode and cathode regions, respectively, in a high-intensity atmospheric-pressure arc discharge [16, 9, 29]. In an ionized gas, the electron mean free path can be approximated as $\lambda_e = 1/(n_e\overline{Q}_{ee} + n_a\overline{Q}_{ea})$, where n_e and n_a are the electron and neutral-atom number densities, respectively, and $\overline{Q}_{ee}$ and $\overline{Q}_{ea}$ are the corresponding momentum-transfer collision cross sections for electron–electron and electron–atom collisions. Electron–electron collisions are Coulomb collisions, for which the averaged momentum-transfer cross section, defined as $\overline{Q}_{ee} = e^4 \ln\Lambda/[12\sqrt{2}\,\pi\varepsilon_0^2(k_B T_e)^2]$ [30]. Here, $\ln\Lambda$ is the Coulomb logarithm, given by $\ln\Lambda = \ln\left(1.24\times10^7 T_e^{3/2} n_e^{-1/2}\right)$, where T_e is in K and n_e is in m^{-3}.

The average electron–neutral momentum-transfer cross section is of the order 0.2×10^{-19} m^2 [31] for the electron temperatures considered. Using the corresponding LTE electron and neutral densities, the resulting electron mean free path is of order $\lambda_e \sim 10^{-6}$ m. The characteristic length scale for electron–heavy-particle energy relaxation, λ_T, is approximated as$\lambda_e/\sqrt{2m_e/M}$, where M is the mass of the heavy species (argon in this case) and m_e is the electron mass. Thus, $\lambda_T \sim 10^{-4}$ m. To estimate the diffusion length scale, where the effect of absorbing electrode surfaces becomes important, we use the expression $\lambda_d = (kT_e/e)/E_z$from Lowke and Tanaka [16] which gives $\lambda_d \sim 10^{-4}$ m for a typical arc discharge.

At the smallest spatial scale, the Debye length calculated from the LTE number densities is of order $\lambda_{De} \sim 10^{-8}$ m, which characterizes the spatial extent over which charge separation and space-charge effects become important. The ion mean free path provides another relevant scale and can be estimated as $\lambda_i = 1/(n_a\overline{Q}_{ia})$, where $\overline{Q}_{ia}$ is the momentum-transfer cross section for ion–atom collisions. For argon, we use the approximate fit $\overline{Q}_{ia} = 2[7\times10^{-10} - 0.6\times10^{-10}\ln(2k_B T_h)]$ [32], where k_B is the Boltzmann constant and T_h is the heavy-particle temperature.Use of this expression gives an ion mean free path of order $\lambda_i \sim 10^{-7}$ m.

The resulting hierarchy of characteristic length scales in the near-electrode region is therefore $\lambda_{De} \ll \lambda_i \ll \lambda_e \ll \lambda_d \sim \lambda_T$. These estimates indicate that the near-electrode region over which significant departures from LTE develop extends over a characteristic length scale of order 10^{-4} m. Other modeling results and analysis shows the same [24, 19, 22, 20]. Consistent with this estimate, Lowke and Tanaka [16] showed that excluding a near-electrode region of order 10^{-4} m is sufficient to model the remaining plasma using an LTE approximation. This separation of scales provides the basis as well a necessity to treat the near-electrode region separately from the bulk plasma.

We therefore model the electrode region as a one-dimensional domain of length 2e-4 m with the electrode surface at one end and the bulk plasma at the other. Externally imposed bulk flows, such as the cathode jet, are not represented. However the bulk fluid is not assumed to be at rest during the transiet stages of the discharge. The region is assumed to consist of three species: electrons (e), singly charged argon ions (i), and argon atoms (n). The presence of multiply charged species is

neglected as justified in [33]. Ions and neutrals exchange momentum and energy rapidly through resonant charge-exchange collisions, whose mean free path λ_i is small compared with the length scale of the temperature variation, and are therefore are treated as a heavy particle mixture with a common temperature, $T_h = T_i = T_n$. The electrons are described by a separate unique temperature T_e, which is justified in Section 3.8.

We therfore solve for species number densities, electron and heavy particle temperatures, and the electric field to describe the plasma in the electrode region. The governing equations and the involved assumptions are described below.

2.1.1 Species continuity The number densities of the three species obey

$$\frac{\partial n_e}{\partial t} + \nabla \cdot \mathbf{\Gamma}_e = S, \qquad \frac{\partial n_i}{\partial t} + \nabla \cdot \mathbf{\Gamma}_i = S, \qquad \frac{\partial n_n}{\partial t} + \nabla \cdot \mathbf{\Gamma}_n = -S, \tag{1}$$

with the net ionization rate

$$S = k_{\mathrm{ion}}(T_e)\, n_e n_n - k_{\mathrm{rec}}(T_e)\, n_e^2 n_i. \tag{2}$$

Here k_{ion} and k_{rec} are the rate coefficients for electron-impact ionization and for its inverse, three-body recombination with an electron as the third body, taken from Hoffert and Lien [34]. These are the only reactions retained. Every ionization event creates one electron and one ion and removes one atom, so the source is common to the electrons and ions and opposite for the neutrals.

In what follows, $p_e = n_e k_B T_e$, $p_i = n_i k_B T_h$, and $p_n = n_n k_B T_h$ are the partial pressures and $p = p_e + p_i + p_n$ is the total pressure. $n_h = n_i + n_n$ and $\rho = M n_h$ are the heavy-particle number and mass densities, with M the argon mass. $\mathbf{u}_s$ is the mean velocity of species s and $\mathbf{\Gamma}_s = n_s \mathbf{u}_s$ its particle flux. $\mathbf{u}$ is the mass-averaged heavy-particle velocity, defined by $\rho \mathbf{u} = M(n_i \mathbf{u}_i + n_n \mathbf{u}_n)$, with the electron mass neglected. These frequencies and the remaining transport coefficients are functions of the local state $(n_e, n_i, n_n, T_e, T_h, |\mathbf{E}|)$ and are given in Appendix 3.

2.1.2 Species momentum balances The momentum equation of species s is $m_s n_s\, D_s \mathbf{u}_s / Dt = \mathbf{F}_s + \mathbf{R}_s$. Here $\mathbf{F}_s$ are all the forces acting on species s. Neglecting the inertia of the species relative to the heavy gas, $D_s \mathbf{u}_s / Dt \simeq D\mathbf{u}/Dt$, inplies each species accelerates with the bulk plasma and there is no relative acceleration between species. That bulk acceleration stems from the heavy-gas momentum equation (18), $\rho\, D\mathbf{u}/Dt = \mathbf{F}$, where $\mathbf{F}$ is the net force per unit volume on the plasma, defined in equation (7) below. Writing $Y_s = m_s n_s / \rho$ for the mass fraction of species s, so that $Y_i = n_i/n_h$, $Y_n = n_n/n_h$, $Y_e = m_e n_e/(M n_h)$, and $Y_e + Y_i + Y_n = 1$, the momentum equations reduce to the local force balances [35]

$$0 = -e n_e \mathbf{E} - \nabla p_e - m_e n_e \bar{\nu}_{en}^M (\mathbf{u}_e - \mathbf{u}_n) - m_e n_e \bar{\nu}_{ei}^M (\mathbf{u}_e - \mathbf{u}_i) - C_e n_e k_B \nabla T_e - Y_e \mathbf{F}, \tag{3}$$

$$0 = +e n_i \mathbf{E} - \nabla p_i - M n_i \bar{\nu}_{in}^M (\mathbf{u}_i - \mathbf{u}_n) - M n_i \bar{\nu}_{ie}^M (\mathbf{u}_i - \mathbf{u}_e) + C_i n_e k_B \nabla T_e - Y_i \mathbf{F}, \tag{4}$$

$$0 = -\nabla p_n - M n_n \bar{\nu}_{ni}^M (\mathbf{u}_n - \mathbf{u}_i) - M n_n \bar{\nu}_{ne}^M (\mathbf{u}_n - \mathbf{u}_e) + C_n n_e k_B \nabla T_e - Y_n \mathbf{F}. \tag{5}$$

The frequencies $\bar{\nu}_{en}^M$, $\bar{\nu}_{ei}^M$, $\bar{\nu}_{in}^M$, $\bar{\nu}_{ie}^M$, $\bar{\nu}_{ni}^M$, and $\bar{\nu}_{ne}^M$ are *averaged momentum-transfer collision frequencies*. The superscript M denotes the average over the velocity distribution that reproduces the mobility (equivalently, the electrical conductivity) of the species. It differs from the energy-exchange averages $\bar{\nu}^E$, which enter the elastic energy transfer in equations (14) and (16). The collision frequencies are defined in (Appendix 3). C_e is the electron thermal-diffusion coefficient, and C_i and C_n are its reactions on the ions and neutrals. Momentum conservation in each collision pair ties the frequencies together,

$$m_e n_e \bar{\nu}_{en}^M = M n_n \bar{\nu}_{ne}^M, \qquad m_e n_e \bar{\nu}_{ei}^M = M n_i \bar{\nu}_{ie}^M, \qquad n_i \bar{\nu}_{in}^M = n_n \bar{\nu}_{ni}^M, \tag{6}$$

and the thermal-diffusion forces sum to zero, $C_i + C_n = C_e$. The force that the electron temperature gradient exerts on the electrons is therefore exactly balanced by its reaction on the two heavy species.

Adding equations (3)–(5) and using equation (6) removes every collisional and thermal-diffusion term. The species' own electric and pressure forces add up to the net force on the plasma,

$$\sum_s \mathbf{F}_s = \mathbf{F} \equiv e(n_i - n_e)\mathbf{E} - \nabla p. \tag{7}$$

Because $\sum_s Y_s = 1$, the remaining terms cancel identically, and the sum reduces to $0 = 0$. The $Y_s \mathbf{F}$ terms are what make the species balances consistent with a heavy gas that accelerates under the

net force (7). Because their sum is an identity, equations (3)–(5) contain only two independent relations, which determine two relative velocities. The third relation is obtained by the definition of the bulk velocity $\mathbf{u}$, and neglecting the electron mass.

$$\mathbf{u} = \frac{n_i \mathbf{u}_i + n_n \mathbf{u}_n}{n_i + n_n} \tag{8}$$

Two of equations (3)–(5) then determine the electron and ion velocities relative to the heavy gas. Equation (8) determines the neutral velocity once the frame velocity $\mathbf{u}$ is known from the heavy-gas momentum equation, equation (18).

2.1.3 Species fluxes *Electrons.* With $\bar{\nu}_e^M = \bar{\nu}_{en}^M + \bar{\nu}_{ei}^M$ the total electron momentum-transfer frequency, solving equation (3) for $\mathbf{u}_e$ and expanding $\nabla p_e = k_B T_e \nabla n_e + n_e k_B \nabla T_e$ gives

$$\mathbf{\Gamma}_e = -\frac{e n_e}{m_e \bar{\nu}_e^M}\mathbf{E} - \frac{k_B T_e}{m_e \bar{\nu}_e^M}\nabla n_e - (1 + C_e)\frac{n_e k_B}{m_e \bar{\nu}_e^M}\nabla T_e + \frac{n_e}{\bar{\nu}_e^M}\left(\bar{\nu}_{en}^M \mathbf{u}_n + \bar{\nu}_{ei}^M \mathbf{u}_i\right). \tag{9}$$

The first two terms are ordinary drift and diffusion. Their coefficients, $e/(m_e\bar{\nu}_e^M)$ and $k_B T_e/(m_e\bar{\nu}_e^M)$, are the magnitudes of the electron mobility and diffusivity, and because both are set by the same frequency they satisfy the Einstein relation identically. The third term combines the temperature-gradient part of the electron pressure force with the additional thermal-diffusion (Soret) force C_e. The last term is the collisional drag exerted on the electrons by the moving heavy species.

Ions. The ion balance (4) contains the friction against the neutrals, $\propto \mathbf{u}_i - \mathbf{u}_n$, and against the electrons, $\propto \mathbf{u}_i - \mathbf{u}_e$. Through the frame condition (8), $\mathbf{u}_n - \mathbf{u} = -(n_i/n_n)(\mathbf{u}_i - \mathbf{u})$, so $\mathbf{u}_i - \mathbf{u}_n = (n_h/n_n)(\mathbf{u}_i - \mathbf{u})$, while $\mathbf{u}_i - \mathbf{u}_e = (\mathbf{u}_i - \mathbf{u}) - (\mathbf{u}_e - \mathbf{u})$. The total friction on the ions is therefore $-M n_i \tilde{\nu}_i^M (\mathbf{u}_i - \mathbf{u}) + M n_i \bar{\nu}_{ie}^M (\mathbf{u}_e - \mathbf{u})$, with the effective ion momentum-transfer frequency

$$\tilde{\nu}_i^M = \frac{n_h}{n_n}\,\bar{\nu}_{in}^M + \bar{\nu}_{ie}^M. \tag{10}$$

Solving equation (4) for $\mathbf{u}_i$ then gives the ion flux

$$\mathbf{\Gamma}_i = \frac{e n_i}{M\tilde{\nu}_i^M}\mathbf{E} - \frac{k_B T_h}{M\tilde{\nu}_i^M}\nabla n_i - \frac{n_i k_B}{M\tilde{\nu}_i^M}\nabla T_h + C_i\frac{n_e k_B}{M\tilde{\nu}_i^M}\nabla T_e - \frac{n_i}{n_h}\frac{\mathbf{F}}{M\tilde{\nu}_i^M} + n_i\left[\mathbf{u} + \frac{\bar{\nu}_{ie}^M}{\tilde{\nu}_i^M}(\mathbf{u}_e - \mathbf{u})\right]. \tag{11}$$

Every ion transport term is governed by the single frequency $\tilde{\nu}_i^M$. Its ion–neutral part is enhanced by the factor $n_h/n_n \geq 1$ over the bare charge-exchange frequency $\bar{\nu}_{in}^M$: an ion drifting relative to the heavy gas is matched, through the frame condition, by a counter-flow of neutrals, which raises the ion–neutral relative velocity. In the weakly ionized limit, $n_i \ll n_n$, $\tilde{\nu}_i^M \to \bar{\nu}_{in}^M + \bar{\nu}_{ie}^M$. The enhancement stays finite as $n_n \to 0$, because $\bar{\nu}_{in}^M \propto n_n$. The third term is ion diffusion driven by the heavy-particle temperature gradient. The fourth is the reaction of the ions to the electron thermal-diffusion force. The fifth is the ions' share, $Y_i\mathbf{F}$, of the net force on the plasma. This share accelerates the ions together with the rest of the heavy gas and so does not drive them relative to it. Relative to the ion's own driving force, it is of order the ionization degree n_i/n_h. In the fully ionized limit ($n_n \to 0$, $Y_i \to 1$) it cancels the ion's own electric and pressure forces, leaving $e n_e \mathbf{E} + \nabla p_e$, the reaction to the force on the electrons. The ion motion relative to the frame is then set entirely by electron–ion friction, as momentum conservation requires. The last term is advection by the heavy gas together with the electron–ion drag.

Neutrals. Multiplying the frame condition (8) by n_n gives the neutral flux directly,

$$\mathbf{\Gamma}_n = n_h \mathbf{u} - \mathbf{\Gamma}_i, \tag{12}$$

so no independent neutral transport coefficient is required. The neutral balance (5) is not needed separately, since it follows from the other two balances and the identity obtained by summing all three. In the absence of heavy-gas motion, $\mathbf{\Gamma}_n = -\mathbf{\Gamma}_i$: every ion carried towards the electrode is returned as a neutral atom.

With the drag, thermal-diffusion, and mixture-acceleration terms removed ($C_e = C_i = 0$, $Y_i\mathbf{F} = 0$, $\mathbf{u}_s = \mathbf{u} = 0$ in the drag terms, $n_i \ll n_n$), equations (9) and (11) reduce to the classical drift-diffusion fluxes, with mobilities $e/(m_e\bar{\nu}_e^M)$ and $e/[M(\bar{\nu}_{in}^M + \bar{\nu}_{ie}^M)]$. Unlike those fluxes, they retain the full pressure-gradient force of each species in a non-isothermal plasma.

2.1.4 Electrostatic field The electric field is obtained from Poisson's equation,

$$\nabla \cdot (\varepsilon_0 \mathbf{E}) = e(n_i - n_e), \qquad \mathbf{E} = -\nabla\phi, \tag{13}$$

which is solved over the entire computational domain. The transition from the quasi-neutral bulk plasma to the space-charge region is thereby captured without imposing a priori sheath boundaries.

2.1.5 Electron energy The electron thermal energy density $\mathcal{E} = \frac{3}{2} n_e k_B T_e$ obeys

$$\frac{\partial \mathcal{E}}{\partial t} + \nabla \cdot \mathbf{h}_e = -e\,\boldsymbol{\Gamma}_e \cdot \mathbf{E} - \frac{3}{2} n_e k_B (T_e - T_h)\,\frac{2m_e}{M}\,\bar{\nu}_{eh}^{E} - \varepsilon_{\text{ion}} S, \tag{14}$$

with the electron energy flux

$$\mathbf{h}_e = \frac{5}{2} k_B T_e\,\boldsymbol{\Gamma}_e - \kappa_e \nabla T_e + k_B T_e\, n_e \Big[C_i(\mathbf{u}_e - \mathbf{u}_i) + C_n(\mathbf{u}_e - \mathbf{u}_n) \Big]. \tag{15}$$

The terms on the right-hand side of equation (14) are Joule heating by the field acting on the total (drift plus diffusion) electron flux, and elastic energy transfer to both heavy species at the averaged energy-exchange frequency $\bar{\nu}_{eh}^{E} = \bar{\nu}_{en}^{E} + \bar{\nu}_{ei}^{E}$. The superscript E marks the energy-exchange average over the electron distribution. It differs from the momentum-transfer (mobility) averages $\bar{\nu}^{M}$ of equations (3)–(5) and is evaluated in Appendix 3. The last term is the ionization-energy sink of the net ionization rate, with $\varepsilon_{\text{ion}} = 15.76$ eV the first ionization potential of argon. Three-body recombination returns the energy that ionization removes. The energy flux (15) is the sum of the enthalpy flux at $\frac{5}{2} k_B T_e$ per electron, Fourier conduction with the zero-current electron thermal conductivity κ_e, and a thermal-diffusion heat flux. The last of these is the Onsager reciprocal of the thermal-diffusion forces in equations (4)–(5) and carries the same coefficients C_i and C_n. Radiation losses are usually small [19, 24] and are not included.

2.1.6 Heavy-particle energy The common heavy-particle temperature is governed by

$$n_h c_v \left(\frac{\partial T_h}{\partial t} + \mathbf{u} \cdot \nabla T_h \right) = \nabla \cdot (\kappa_h \nabla T_h) + \frac{3}{2} n_e k_B (T_e - T_h)\,\frac{2m_e}{M}\,\bar{\nu}_{eh}^{E} + Q_{in} - n_h k_B T_h\, \nabla \cdot \mathbf{u}, \tag{16}$$

where $c_v = \frac{3}{2} k_B$ is the translational heat capacity per heavy particle and κ_h is the heavy-particle thermal conductivity (Appendix 3). The elastic-exchange term is the exact negative of the electron loss in equation (14), with the same $\bar{\nu}_{eh}^{E}$ and the same local values, so the heavy gas gains exactly the energy the electrons lose. The ionization energy is charged entirely to the electrons and does not appear here. The ion–neutral frictional heating is

$$Q_{in} = M n_i\, \bar{\nu}_{in}^{M}\, |\mathbf{u}_i - \mathbf{u}_n|^2, \tag{17}$$

which is positive definite and reduces to the Joule dissipation of the ion drift when the neutral gas is at rest. It differs from $\mathbf{E} \cdot \mathbf{j}_i$, which can be negative in a strong sheath field where the field does work against ions diffusing up their own density gradient. The last term in equation (16) is the compressive work of the heavy gas.

2.1.7 Heavy-gas momentum The mass-averaged heavy-gas velocity is advanced by an inviscid single-fluid momentum equation,

$$\frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u}\mathbf{u}) = -\nabla p + e(n_i - n_e)\mathbf{E}, \qquad p = n_e k_B T_e + n_h k_B T_h, \tag{18}$$

that is, $\rho\, D\mathbf{u}/Dt = \mathbf{F}$, with $\mathbf{F}$ the net force (7); the conservative form follows from heavy-particle mass conservation, shown below. This equation governs the motion of the heavy gas as a whole. It also supplies the mixture acceleration whose shares $Y_s\mathbf{F}$ enter the species balances (3)–(5), which in turn govern only the motion of each species relative to the heavy gas. Because electron inertia is negligible, the electric and pressure forces on the electrons are transmitted to the heavy gas by friction, so both the electron pressure and the net space-charge force appear on the right-hand side. Together with the neutral continuity equation and the flux (12), equation (18) implies $\partial n_h/\partial t + \nabla \cdot (n_h \mathbf{u}) = 0$: heavy-particle mass is conserved exactly, since ionization only converts atoms into ions. The total pressure relaxes acoustically rather than instantaneously. If the inertial terms are dropped, $\mathbf{F} = 0$: the $Y_s\mathbf{F}$ terms vanish, and the algebraic mechanical-equilibrium constraint $\nabla p = e(n_i - n_e)\mathbf{E}$ is recovered, which is the limit underlying constant-pressure (Dalton's-law) closures for the neutral density. Viscous stresses are not included.

The state advanced in time is therefore $\{n_e, n_i, n_n, T_e, T_h, \mathbf{u}, \phi\}$. It is governed by the continuity equations (1)–(2) with the fluxes (9), (11), and (12); Poisson's equation (13); the electron and heavy-particle energy equations (14)–(17); and the heavy-gas momentum equation (18).

2.2 Boundary conditions

The system is closed by conditions at the two ends of the domain: the electrode surface, $x = 0$, and the edge of the near-electrode layer where the solution is matched to the adjacent LTE bulk plasma, $x = L$. Below, $\hat{\mathbf{n}}$ denotes the unit normal at the electrode pointing into the wall.

At the electrode, the surface is taken to be absorbing and non-emitting, and is held at the wall temperature, $T_h(0) = T_{\text{wall}}$, with the potential referenced to zero, $\phi(0) = 0$. Electrons are removed at the one-sided (Maxwellian) thermal-effusion flux,

$$\boldsymbol{\Gamma}_e \cdot \hat{\mathbf{n}} = \tfrac{1}{4} n_e \bar{c}_e, \qquad \bar{c}_e = \sqrt{\frac{8 k_B T_e}{\pi m_e}}, \tag{19}$$

and each absorbed electron removes a mean energy of $2k_B T_e$, so that $\mathbf{h}_e \cdot \hat{\mathbf{n}} = 2k_B T_e \, \boldsymbol{\Gamma}_e \cdot \hat{\mathbf{n}}$. Ions are collected at their thermal-effusion flux augmented by any drift directed into the wall,

$$\boldsymbol{\Gamma}_i \cdot \hat{\mathbf{n}} = \tfrac{1}{4} n_i \bar{c}_i + \max\Big(\frac{e n_i}{M \tilde{\nu}_i^M} \mathbf{E} \cdot \hat{\mathbf{n}}, \, 0 \Big), \qquad \bar{c}_i = \sqrt{\frac{8 k_B T_h}{\pi M}}, \tag{20}$$

which reduces to the thermal flux when the local field does not drive ions into the electrode. This treatment is consistent with the field-aware treatment of the counter-diffusing sheath-edge current used elsewhere [23, 29]. The wall is impermeable to the heavy gas, $\mathbf{u} \cdot \hat{\mathbf{n}} = 0$, so equation (12) gives

$$\boldsymbol{\Gamma}_n \cdot \hat{\mathbf{n}} = -\boldsymbol{\Gamma}_i \cdot \hat{\mathbf{n}} : \tag{21}$$

every ion neutralized at the surface re-enters the plasma as an atom. Thermionic and secondary electron emission are not included in equation (19) at present (Section 4.3), so the electron supply at a hot thermionic cathode is expected to be under-predicted.

At the outer edge, $x = L$, the electron and ion densities and both temperatures are set to Dirichlet values taken from the adjacent LTE bulk solution: $n_e = n_e^{\text{bulk}}$, $n_i = n_i^{\text{bulk}}$, $T_e = T_e^{\text{bulk}}$, and $T_h = T_{\text{bulk}}$. The neutral density follows from Dalton's law at the background pressure p_0,

$$n_n = \frac{p_0 - n_e k_B T_e}{k_B T_h} - n_i, \tag{22}$$

and the heavy-gas velocity is given a zero normal gradient. The arc current is imposed through the electric field: the total current density at the boundary, conduction plus displacement, is held equal to the applied current density J_{app},

$$e \, (\boldsymbol{\Gamma}_i - \boldsymbol{\Gamma}_e) \cdot \hat{\mathbf{e}}_x + \varepsilon_0 \frac{\partial E_x}{\partial t} = J_{\text{app}}, \tag{23}$$

which is integrated in time as an evolution equation for the boundary field. The initial field enters only as the initial condition of this equation.

3 Appendix A. Transport Coefficients

This appendix gives the transport and exchange coefficients that close the governing equations of Section 2.1: the electron momentum-transfer and energy-exchange collision frequencies, thermal-diffusion coefficients, and thermal conductivity; the ion momentum-transfer frequencies; and the heavy-particle heat capacity and thermal conductivity. Every coefficient is a closed-form or tabulated function of the local state.

3.1 Electron transport: general framework

The electron current density $\mathbf{J}_e = -e\boldsymbol{\Gamma}_e$ and heat flux $\mathbf{q}_e$ obey a linear-response (Onsager) relation with two driving forces. These are the generalized field $\boldsymbol{\mathcal{E}} = \mathbf{E} + \nabla p_e / (e n_e)$ and the temperature gradient:

$$\mathbf{J}_e = \sigma \, \boldsymbol{\mathcal{E}} + \varphi \, \nabla T_e, \tag{24}$$

$$\mathbf{q}_e = -\frac{5}{2} \frac{k_B T_e}{e} \mathbf{J}_e - \lambda' \nabla T_e - T_e \varphi \, \boldsymbol{\mathcal{E}} = -\lambda \nabla T_e + \psi \, \mathbf{J}_e, \tag{25}$$

with

$$\lambda = \lambda' - T_e \frac{\varphi^2}{\sigma}, \qquad \psi = -\frac{5}{2} \frac{k_B T_e}{e} - T_e \frac{\varphi}{\sigma}. \tag{26}$$

Here σ is the electrical conductivity, φ the thermal-diffusion coefficient, λ' the thermal conductivity at fixed $\boldsymbol{\mathcal{E}}$, and λ the ordinary (zero-current) thermal conductivity. Comparing equations (24)–(26) with the electron momentum balance (3) and the energy flux (15) identifies

$$\bar{\nu}_e^M = \frac{n_e e^2}{m_e \sigma}, \qquad C_e = \frac{e}{k_B}\frac{\varphi}{\sigma}, \qquad \kappa_e = \lambda, \qquad \frac{5}{2} + C_e = -\frac{e\psi}{k_B T_e}. \tag{27}$$

The conductivity entering the energy equation is the zero-current value λ, not λ'. A temperature gradient alone drives a current $\varphi\nabla T_e$, and holding $\mathbf{J}_e = 0$ requires a field $\boldsymbol{\mathcal{E}} = -(\varphi/\sigma)\nabla T_e$. The heat flux carried by this field reduces the apparent conduction from λ' to λ. Because equation (14) accounts separately for Joule heating and the enthalpy flux, its conductivity must be the zero-current one. The coefficients σ, φ, and λ are evaluated in the partially and fully ionized limits and then bridged.

3.2 Partially ionized limit: electron–neutral transport

When electron–neutral collisions dominate, the electrons form a Lorentz gas. The isotropic part of the distribution is a Maxwellian $f_M(T_e)$, and only the first-order anisotropic correction responds to the field and the temperature gradient. With C the electron speed, $x = m_e C^2/(2k_B T_e)$, and $\nu_{eH}(C) = n_n C Q^{(1)}(\varepsilon)$ the energy-dependent electron–neutral momentum-transfer frequency built from the momentum-transfer cross section $Q^{(1)}(\varepsilon)$, a two-term solution of the electron Boltzmann equation gives the anisotropic correction

$$f^1 = \frac{e}{m_e \nu_{eH}}\frac{\partial f_M}{\partial C}\boldsymbol{\mathcal{E}} - \frac{C}{\nu_{eH}} f_M \left[\nabla \ln p_e + \left(x - \tfrac{5}{2}\right)\nabla \ln T_e\right]. \tag{28}$$

Two different kinds of velocity average appear below, and they are given different notation. The first is the plain average over the Maxwellian, normalized to unity,

$$\langle g \rangle_0 = \int g\, f_M\, d^3C = 4\pi \int_0^\infty g\, C^2 f_M\, dC. \tag{29}$$

The second is the *transport average*. Every current and conduction heat-flux moment of equation (28) carries, besides the Maxwellian, an extra factor C^2/ν_{eH}, so the transport coefficients are expressed through averages under the weight $C^4 f_M/\nu_{eH}$:

$$\langle g \rangle_\nu = \frac{\langle g\, C^2/\nu_{eH}\rangle_0}{\langle C^2/\nu_{eH}\rangle_0} = \frac{\int_0^\infty g\, C^4 f_M \nu_{eH}^{-1}\, dC}{\int_0^\infty C^4 f_M \nu_{eH}^{-1}\, dC}. \tag{30}$$

In energy, this weight is $\varepsilon\, e^{-\varepsilon/k_B T_e}/Q^{(1)}(\varepsilon)$. The transport average is used only for the dimensionless moments $\langle x \rangle_\nu$ and $\langle x^2 \rangle_\nu$ in equation (33) below. It is *not* the average that defines the collision frequencies.

The collision frequencies in the model are Maxwellian averages $\langle\cdot\rangle_0$ of $\nu_{eH}(C)$, and two different ones are needed. Both use the same weight, $C^2 f_M$, that is, the Maxwellian weighted by the electron kinetic energy. They differ in whether ν_{eH} or $1/\nu_{eH}$ is averaged. The *momentum-transfer (mobility) average*, denoted by the superscript M, is the harmonic mean of ν_{eH} under this weight. It reproduces the mobility and is the frequency used in the force balances (3)–(5):

$$\bar{\nu}_{en}^M = \frac{\langle C^2\rangle_0}{\langle C^2/\nu_{eH}\rangle_0} = n_n \frac{3k_B T_e/m_e}{4\pi \int_0^\infty C^3 f_M\, [Q^{(1)}]^{-1}\, dC} = n_n \sqrt{\frac{2}{m_e}}\,\frac{3\sqrt{\pi}}{4}\,\frac{(k_B T_e)^{5/2}}{\int_0^\infty \varepsilon\, e^{-\varepsilon/k_B T_e}\, [Q^{(1)}(\varepsilon)]^{-1}\, d\varepsilon}. \tag{31}$$

The second form follows from $\nu_{eH} = n_n C Q^{(1)}$ and $\langle C^2\rangle_0 = 3k_B T_e/m_e$. The third follows from the change of variable $\varepsilon = \frac{1}{2} m_e C^2$, with $f_M = (m_e/2\pi k_B T_e)^{3/2} e^{-\varepsilon/k_B T_e}$. For a constant cross section, it reduces to $\bar{\nu}_{en}^M = (9\pi/32)\,\frac{4}{3} n_n \bar{c}_e Q^{(1)} \approx 0.88 \times \frac{4}{3} n_n \bar{c}_e Q^{(1)}$. Equation (31) is identical to defining $\bar{\nu}_{en}^M$ through the Lorentz conductivity of [30] (their equation VIII 2.14a), $\sigma_{ea} = n_e e^2/(m_e \bar{\nu}_{en}^M)$. This is how it is evaluated in practice. The *energy-exchange average*, denoted by the superscript E, is the arithmetic mean of ν_{eH} under the same weight:

$$\bar{\nu}_{en}^E = \frac{\langle C^2 \nu_{eH}\rangle_0}{\langle C^2\rangle_0} = \frac{4}{3} n_n \bar{c}_e\, \overline{Q}_{ea}^{(1,1)}(T_e), \qquad \bar{c}_e = \sqrt{\frac{8k_B T_e}{\pi m_e}}. \tag{32}$$

Because a harmonic mean never exceeds the arithmetic mean over the same weight, $\bar{\nu}_{en}^M \le \bar{\nu}_{en}^E$, with equality only for an energy-independent ν_{eH}. The denominator of equation (31) is also the

normalizing integral of the transport average (30). With $\langle C^2\rangle_0 = 3k_BT_e/m_e$, the Lorentz-gas coefficients are

$$\sigma_{ea} = \frac{n_e e^2}{m_e \bar{\nu}_{en}^M}, \qquad \left(\frac{\varphi}{\sigma}\right)_{ea} = \frac{k_B}{e}\left(\langle x\rangle_\nu - \frac{5}{2}\right), \qquad \lambda_{ea} = \frac{n_e k_B^2 T_e}{m_e \bar{\nu}_{en}^M}\left[\langle x^2\rangle_\nu - \langle x\rangle_\nu^2\right]. \tag{33}$$

The electron–neutral thermal-diffusion coefficient $(e/k_B)(\varphi/\sigma)_{ea} = \langle x\rangle_\nu - \frac{5}{2}$ vanishes for an energy-independent collision frequency. Its sign and magnitude therefore reflect the energy dependence of $Q^{(1)}$, which for argon is dominated by the Ramsauer minimum.

The energy-exchange average (32) is the one that gives the elastic energy-loss rate of a Maxwellian electron population, $\frac{2m_e}{M}\langle \varepsilon\, \nu_{eH}\rangle_0(1 - T_h/T_e) = \frac{3}{2}k_B(T_e - T_h)\frac{2m_e}{M}\bar{\nu}_{en}^E$. For argon between 0.5 and 2 eV, $\bar{\nu}_{en}^E$ exceeds $\bar{\nu}_{en}^M$ by a factor of 4–4.5, because energy exchange is dominated by fast electrons above the Ramsauer minimum, while the mobility is dominated by slow electrons trapped inside it. Using $\bar{\nu}_{en}^E$ in the force balances would therefore overestimate the electron–neutral friction by this factor, and using $\bar{\nu}_{en}^M$ in the energy equations would underestimate the elastic loss by the same factor. Equations (30)–(32) are evaluated on the argon momentum-transfer cross section [31] and tabulated as functions of T_e, in reduced form per unit neutral density.

3.3 Fully ionized limit and the Coulomb logarithm

When electron–ion collisions dominate, the Spitzer–Härm solution for a singly charged ion gives [30]

$$\sigma_{ei} = 1.975\,\frac{n_e e^2}{m_e \bar{\nu}_{ei}^E}, \qquad \lambda_{ei} = 3.203\,\frac{n_e k_B^2 T_e}{m_e \bar{\nu}_{ei}^E}, \qquad \varphi_{ei} = 1.389\,\frac{n_e e k_B}{m_e \bar{\nu}_{ei}^E}, \qquad \psi_{ei} = -3.203\,\frac{k_B T_e}{e}. \tag{34}$$

The collision frequency cancels in $(e/k_B)(\varphi/\sigma)_{ei} = 1.389/1.975 = 0.7033$, and the consistency condition of equation (27), $-e\psi/(k_BT_e) = 3.203 = \frac{5}{2} + 0.703$, is satisfied. Here $\bar{\nu}_{ei}^E$ is the energy-exchange (C^2-weighted) average of the electron–ion momentum-transfer frequency,

$$\bar{\nu}_{ei}^E = n_i\,\frac{4\sqrt{2\pi}}{3}\left(\frac{m_e}{k_BT_e}\right)^{3/2}\left(\frac{e^2}{4\pi\varepsilon_0 m_e}\right)^2 \ln\Lambda = 2.906\times10^{-12}\,\frac{n_i[\mathrm{m}^{-3}]\,\ln\Lambda}{T_e[\mathrm{eV}]^{3/2}}\ \mathrm{s}^{-1}, \tag{35}$$

with the Coulomb logarithm

$$\ln\Lambda = \max\left(2,\ \ln\frac{\lambda_{De}}{b_{90}}\right), \qquad \lambda_{De} = \sqrt{\frac{\varepsilon_0 T_e[\mathrm{eV}]}{n_e e}}, \qquad b_{90} = \frac{e}{12\pi\varepsilon_0 T_e[\mathrm{eV}]}. \tag{36}$$

Here b_{90} is the classical 90°-deflection impact parameter. The logarithm is floored at 2 to keep $\bar{\nu}_{ei}^E$ finite when the plasma is not weakly coupled. The rate (35) is set by the ion density, which is the collision partner, whereas the Debye length uses the electron density. The two coincide only under quasi-neutrality, which does not hold inside the sheath.

3.4 Bridging the two limits

At intermediate ionization, the electron–neutral and electron–ion contributions are combined by additive (resistance-type) mixture rules. These rules are exact in both limits and are weighted by the ratio of the energy-exchange averages of the two collision frequencies,

$$P = \frac{\bar{\nu}_{en}^E}{\bar{\nu}_{ei}^E}. \tag{37}$$

Momentum-transfer frequency. The electron–ion momentum-transfer average is $\bar{\nu}_{ei}^M = 0.5\,\bar{\nu}_{ei}^E$, where the factor $0.5 \approx 1/1.975$ is the strongly ionized conductivity weight of equation (34). Adding the electron–neutral and electron–ion resistivities is the same as adding the two momentum-transfer frequencies,

$$\bar{\nu}_e^M = \bar{\nu}_{en}^M + \bar{\nu}_{ei}^M, \qquad \sigma = \frac{n_e e^2}{m_e \bar{\nu}_e^M}, \tag{38}$$

with $\bar{\nu}_{en}^M$ from equation (31). This is the total frequency in the electron flux (9). Relative to the electron–neutral value alone, the electron–ion (Coulomb) contribution reduces the conductivity by the factor $\bar{\nu}_{en}^M/\bar{\nu}_e^M$.

Thermal-diffusion coefficients. The two values of φ/σ are combined as a weighted mean in $1/P$:

$$\left(\frac{\varphi}{\sigma}\right)_e = \frac{(\varphi/\sigma)_{ei}/P + (\varphi/\sigma)_{ea}}{1 + 1/P}, \qquad \left(\frac{\varphi}{\sigma}\right)_{ei} = \frac{1.389}{1.975}\frac{k_B}{e}, \tag{39}$$

so that

$$C_e = \frac{e}{k_B}\left(\frac{\varphi}{\sigma}\right)_e, \qquad C_i = \frac{e}{k_B}\frac{(\varphi/\sigma)_{ei}/P}{1+1/P}, \qquad C_n = C_e - C_i. \tag{40}$$

Only electron–ion collisions couple momentum directly between electrons and ions. The ion reaction C_i is therefore the electron–ion (Coulomb) part of the thermal-diffusion force alone. The electron–neutral part acts on the atoms, giving $C_n = C_e - C_i$, and the sum rule $C_i + C_n = C_e$ used in Section 2.1 holds identically.

Thermal conductivity. The zero-current conductivities combine as parallel thermal resistances,

$$\frac{1}{\lambda_e} = \frac{1}{\lambda_{ea}} + \frac{1}{\lambda_{ei}}, \qquad \kappa_e = \lambda_e \equiv K\,\frac{n_e k_B^2 T_e}{m_e \bar{\nu}_e^M}, \tag{41}$$

with $\bar{\nu}_e^M$ the total momentum-transfer frequency of equation (38). Both λ_{ea} and λ_{ei} scale linearly with n_e, so the dimensionless factor K is independent of the electron density. It tends to $\langle x^2\rangle_\nu - \langle x\rangle_\nu^2$ in the electron–neutral limit and to $3.203\,\bar{\nu}_{ei}^M/\bar{\nu}_{ei}^E = 3.203/2 \approx 1.60$ in the fully ionized limit.

Elastic exchange frequency. The total elastic energy-exchange frequency in equations (14) and (16) is the sum of the two energy-exchange averages, not of the momentum-transfer averages $\bar{\nu}^M$:

$$\bar{\nu}_{eh}^E = \bar{\nu}_{en}^E + \bar{\nu}_{ei}^E. \tag{42}$$

Both terms are, by construction, the elastic energy-loss rate of a near-Maxwellian electron population. The energy lost by the electrons in equation (14) is therefore exactly the energy gained by the heavy gas in equation (16).

3.5 Ion collision frequencies

The ion–neutral transport is governed by resonant charge exchange. The momentum-transfer cross section is taken from the fit [32]

$$\overline{Q}_{ia} = 2\left[7\times10^{-10} - 0.6\times10^{-10}\ln\left(\frac{2k_BT_h}{e}\right)\right]^2 \ \mathrm{m}^2, \tag{43}$$

with $2k_BT_h/e$ in eV. The corresponding low-field mobility is

$$\mu_i^0 = \frac{1.104\times10^5}{\overline{Q}_{ia}\sqrt{T_h}\,n_n} \ \mathrm{m^2\,V^{-1}\,s^{-1}} \qquad (\overline{Q}_{ia}\text{ in m}^2,\ T_h\text{ in K},\ n_n\text{ in m}^{-3}). \tag{44}$$

The high reduced fields near the electrode heat the ions. This is accounted for by a closed-form two-temperature Wannier formulation [36], in which the drift speed v_d and the effective ion temperature T_{eff} are determined self-consistently:

$$v_d^2 = \frac{\sqrt{1+4\mathcal{A}\mathcal{B}}-1}{2\mathcal{B}}, \qquad \mathcal{A} = \left(\mu_i^0|\mathbf{E}|\right)^2, \qquad \mathcal{B} = \frac{M}{3k_BT_h}, \tag{45}$$

$$T_{\mathrm{eff}} = T_h + \frac{Mv_d^2}{3k_B}, \qquad \mu_{in} = \mu_i^0\sqrt{\frac{T_h}{T_{\mathrm{eff}}}}. \tag{46}$$

Equation (45) is the positive root of the quadratic in v_d^2 obtained by substituting $\mu_{in}(T_{\mathrm{eff}})$ into $v_d = \mu_{in}|\mathbf{E}|$. In the low-field limit, $v_d \to \mu_i^0|\mathbf{E}|$, $T_{\mathrm{eff}} \to T_h$, and $\mu_{in} \to \mu_i^0$. The charge-exchange data are available in this mobility form. They enter the model only through the ion–neutral momentum-transfer frequency

$$\bar{\nu}_{in}^M = \frac{e}{M\mu_{in}}. \tag{47}$$

The ion–electron frequency follows from momentum conservation, equation (6),

$$\bar{\nu}_{ie}^M = \frac{m_e n_e}{M n_i}\,\bar{\nu}_{ei}^M, \tag{48}$$

with $\bar{\nu}_{ei}^M = 0.5\,\bar{\nu}_{ei}^E$ as in equation (38). Equations (47) and (48) give the effective ion frequency $\tilde{\nu}_i^M = (n_h/n_n)\bar{\nu}_{in}^M + \bar{\nu}_{ie}^M$ of equation (10), which sets every ion transport term in equation (11). In the weakly ionized bulk, $\bar{\nu}_{ie}^M \ll \bar{\nu}_{in}^M$ and $n_i \ll n_n$, so $\tilde{\nu}_i^M \to \bar{\nu}_{in}^M$. As $n_n \to 0$ (near-total ionization), $\tilde{\nu}_i^M$ stays finite, because $\bar{\nu}_{in}^M \propto n_n$, and is bounded below by the Coulomb-collision value $\bar{\nu}_{ie}^M$. The ion–neutral frictional heating (17) uses the bare charge-exchange frequency $\bar{\nu}_{in}^M$ of equation (47), not $\tilde{\nu}_i^M$, which also contains the counter-flow and electron-friction contributions.

3.6 Heavy-particle heat capacity and thermal conductivity

The heat capacity per heavy particle in equation (16) contains only the translational degrees of freedom, $c_v = \frac{3}{2}k_B$. The gas is compressible, and the work of expansion appears explicitly as $-n_h k_B T_h \nabla\cdot\mathbf{u}$. The electron energy is carried separately, and the ionization energy enters as a source in the electron energy equation. The heavy-particle thermal conductivity is the sum of an atom and an ion channel, each given by a first-approximation Chapman–Enskog expression:

$$\kappa_h = \kappa_a + \kappa_i, \tag{49}$$

$$\kappa_a = \frac{75k_B}{64}\sqrt{\frac{\pi k_B T_h}{M}}\,\frac{n_n}{n_n\overline{Q}_{aa} + n_i\overline{Q}_{ia}}, \qquad \kappa_i = \frac{75k_B}{64}\sqrt{\frac{\pi k_B T_h}{M}}\,\frac{n_i}{n_n\overline{Q}_{ia} + n_i\overline{Q}_{ii}}, \tag{50}$$

with the viscosity-type $(2,2)$ cross sections

$$\overline{Q}_{aa}^{(2,2)} = 1.12\times10^{-18}\,T_h^{-0.2}, \qquad \overline{Q}_{ia}^{(2,2)} = 3.6\times10^{-18}\,T_h^{-0.3}, \qquad \overline{Q}_{ii}^{(2,2)} = \frac{e^4\ln\Lambda}{36\pi\varepsilon_0^2(k_BT_h)^2}, \tag{51}$$

in m^2 with T_h in K. Here $\ln\Lambda$ is the Coulomb logarithm of equation (36). The expression is regular in both limits. As $n_n \to 0$, κ_i tends to the pure ion–ion Coulomb value, and as $n_i \to 0$, κ_a tends to the atom–atom value of the neutral gas.

3.7 Reaction rate coefficients

The ionization and three-body recombination rate coefficients $k_{\mathrm{ion}}(T_e)$ and $k_{\mathrm{rec}}(T_e)$ in equation (2) are effective rate coefficients from Hoffert and Lien [34]. They are tabulated in T_e and interpolated log-linearly. The energy charged to the electrons per net ionization event is $\varepsilon_{\mathrm{ion}} = 15.76$ eV.

3.8 Quasi-Maxwellian electrons near the electrode

Even where the heavy species are strongly out of equilibrium, the electron population can remain closely Maxwellian provided electron–electron and electron–ion collisions are frequent enough to redistribute directed (drift) energy into the isotropic thermal spread faster than electron–heavy collisions or the imposed field can distort it. Following Dinulescu and Pfender's analysis of the anode boundary layer [19], two conditions must hold. First, electron–electron/ion thermalization must dominate electron–neutral momentum transfer, weighted by the mass ratio that makes each electron–neutral collision inefficient at removing energy:

$$\sum_h \frac{m_e}{m_h}\,\frac{\bar\nu_{eh}}{\bar\nu_{ee}} \ll 1. \tag{52}$$

Second, the ordered drift energy imparted by the field must remain small compared with the electron thermal energy, weighted by the same mass ratio:

$$\left(\frac{U_e}{\bar c_e}\right)^2 \ll \frac{m_e}{m_h}. \tag{53}$$

For a representative near-anode condition, $T_e \sim 1$ eV, $n_e \sim 5\times10^{22}\,\mathrm{m}^{-3}$, neutral density $N \sim 6\times10^{23}\,\mathrm{m}^{-3}$, electron–atom momentum-transfer cross section $\bar Q_{ea} = 2\times10^{-20}\,\mathrm{m}^2$, and mean thermal speed $\bar c_e = 7\times10^5$ m/s: the electron–ion/electron–electron frequency from the NRL formulary (with $\ln\Lambda \approx 3$) is $\nu_{ee} \sim \nu_{ei} \sim 5\times10^{11}\,\mathrm{s}^{-1}$, while the electron–neutral frequency is $\nu_{en} = \bar c_e N \bar Q_{ea} \approx 8\times10^9\,\mathrm{s}^{-1}$, i.e. $\nu_{en}/\nu_{ee} \sim 0.016$; with $m_e/M_{\mathrm{Ar}} \approx 1.4\times10^{-5}$, condition (52) gives $\sim 2\times10^{-7} \ll 1$. For condition (53), the electrical conductivity of the layer, $\sigma \approx 5000$ S/m, corresponds to an electron momentum-transfer frequency $\bar\nu_e^M = e^2 n_e/(m_e\sigma) \approx 3\times10^{11}\,\mathrm{s}^{-1}$; with a representative current density $J \sim 10^7\,\mathrm{A/m}^2$, the electron drift speed is $U_e = J/(en_e) \approx 1.3\times10^3$ m/s, so $(U_e/\bar c_e)^2 \approx 3\times10^{-6}$, again well below $m_e/M_{\mathrm{Ar}} \approx 1.4\times10^{-5}$. Both conditions are comfortably satisfied, so the electron population is treated as quasi-Maxwellian throughout the near-electrode layer, and a single electron temperature T_e (rather than a fully kinetic electron energy distribution) is retained in equation (14) there as well as in the bulk.

3.9 Length scales and the thickness of the near-electrode layer

The same parameters fix a hierarchy of length scales that separates the space-charge sheath from the broader near-electrode non-equilibrium region that must be resolved around it. The electron Debye length,

$$\lambda_{De} = \sqrt{\frac{\varepsilon_0 T_e}{n_e e}} \approx 3\times10^{-8}\ \mathrm{m}, \tag{54}$$

sets the scale of the space-charge sheath itself. The electron mean free path against the combined electron–neutral and electron–electron/ion collision frequency used above,

$$\lambda_e = \frac{\bar{c}_e}{\nu_{en} + \nu_{ee}} \approx 1 \times 10^{-6}\ \mathrm{m}, \tag{55}$$

is ~ 30 times λ_{De}, confirming that the sheath is thin compared with a mean free path and can support the non-collisional electron kinetics discussed in Section 3.10. The electron–heavy energy-relaxation length, over which an electron transfers its excess thermal energy to the heavy gas through the inefficient mass-ratio-weighted elastic channel in equation (14), is larger still,

$$\lambda_T = \frac{\lambda_e}{\sqrt{2m_e/M}} \approx 2 \times 10^{-4}\ \mathrm{m}. \tag{56}$$

Finally, the diffusion length that controls how far the electron/ion density profile itself can depart from its local equilibrium value under a given field, used directly by Lowke and Tanaka's mesh-sizing rule [16], is $\Lambda = (kT_e/e)/E$; with $T_e \approx 1$ V and a field $E = J/\sigma \approx (10^7\ \mathrm{A/m^2})/(5000\,\mathrm{S/m}) = 2000$ V/m consistent with the parameters above,

$$\Lambda \approx 5 \times 10^{-4}\ \mathrm{m}. \tag{57}$$

The four scales satisfy $\lambda_{De} \ll \lambda_e \ll \Lambda \sim \lambda_T$, the same ordering (Debye length $\ll$ mean free path $\ll$ boundary-layer thickness) reported directly in Table I of [19] for the anode boundary layer of a comparable arc, $\lambda_{De} \sim (2\text{–}3) \times 10^{-8}$ m, $\lambda_e \sim (1\text{–}2) \times 10^{-6}$ m, boundary-layer thickness $L \sim 0.2$–0.6 mm. On this basis we take a near-electrode layer of thickness $\delta \approx 0.2$ mm — comparable to both Λ and λ_T, and consistent with the range of L found in [19] — as the region over which the unified, non-equilibrium model of Section 4 must be applied; outside it, the plasma is close enough to collision-dominated equilibrium for the LTE bulk treatment of Section to hold. Note that this ∼0.2 mm layer is not itself the space-charge sheath: it is the outer, thermal/diffusion-dominated non-equilibrium region (corresponding to layers 2–3 of the four-sublayer structure found by [24]) inside which a much thinner space-charge sheath, of order λ_{De} to a few times λ_{De}, is nested.

3.10 Is the sheath collisionless?

A collisionless sheath is the standard simplifying assumption used to close the NLTE-sheath and two-temperature ionization-layer-sheath approaches of Section 1.1, and it is tempting to expect it to hold automatically at atmospheric pressure, where collision rates are high. Almeida, Benilov and Naidis show this expectation is wrong: computing the number of ion collisions actually incurred while crossing the sheath gives values from order unity to 10^4, not $\ll 1$, even for atmospheric-pressure argon [24]. The reason is that the sheath thickness itself scales with the *local* Debye length, $\lambda_{De} \propto n_e^{-1/2}$, which is set by the arc's own high electron density and shrinks faster, as the arc becomes hotter and more strongly ionized, than the neutral mean free paths shrink with pressure (the Ramsauer minimum in the argon electron–neutral cross section keeps λ_e from falling as quickly as λ_{De}). A collisionless sheath at high pressure is therefore not evidence of a crude or incomplete model; it is close to the generic outcome for a dense atmospheric-pressure arc, and the question that actually needs answering species by species and quantity by quantity is *how* collisionless, and for what.

We make this quantitative with four Knudsen numbers, each the ratio of the relevant mean free path to the resolved sheath width s: $\mathrm{Kn}_e = \lambda_e/s$ (do electrons collide crossing the sheath at all); $\mathrm{Kn}_\varepsilon = \lambda_T/s$ (electron–heavy energy exchange: does a local electron energy balance exist inside the sheath); $\mathrm{Kn}_{ee} = \lambda_{ee}/s$ (electron–electron thermalization: does a well-defined T_e, i.e. a locally Maxwellian electron energy distribution, exist inside the sheath at all – a distinct and much more restrictive question than Kn_ε, since λ_{ee} and λ_T differ by roughly 10^4); and $\mathrm{Kn}_i = \lambda_i/s$ for the ion charge-exchange mean free path. For the converged, resolved-sheath operating point of the PLASMEX test case discussed in Section 4 ($s = 1.65 \times 10^{-7}$ m $= 5.6\,\lambda_{De}$, $T_e \approx 0.98$ eV at the wall), these evaluate to $\mathrm{Kn}_e \approx 63$, $\mathrm{Kn}_\varepsilon \approx 1.2 \times 10^4$, $\mathrm{Kn}_{ee} \approx 5.3$, and $\mathrm{Kn}_i \approx 0.8$–$1.7$: the sheath is collisionless for electron momentum and electron–heavy energy exchange by a wide margin, only marginally collisionless for electron thermalization, and carries of order one ion collision, consistent with the lower end of the 10^2–10^4 range quoted by [24] for a somewhat different case. Despite $\mathrm{Kn}_{ee} \sim 5$ being only marginal, an electron population entering a purely retarding, elastic potential barrier is mapped onto itself by Liouville's theorem: reflection by an energy-conserving barrier preserves a Maxwellian at unchanged T_e while depleting the density by the Boltzmann factor,

giving a testable, parameter-free reference prediction, $T_e(x) = T_{e,\mathrm{edge}} = \text{const}$ across the sheath, against which the local closure used in equation (14) can be checked directly (Section 4). No such collisionless reference state exists for the ions, which do not see a retarding barrier and for which $\mathrm{Kn}_i = O(1)$ rules out both the collisionless limit and full collisional equilibrium; ion behaviour in the sheath is therefore treated as an explicit target for the resolved drift-diffusion solution rather than assumed a priori in either limit.

4 Unified near-electrode model and numerical implementation

4.1 Resolving the sheath directly, rather than matching or homogenizing it

Section 1.1 distinguished three published strategies for coupling a bulk-plasma closure to the electrode: solving one boundary value problem with Poisson's equation over the whole gap [24]; matching a non-equilibrium bulk solution directly to an analytic sheath closure at an interface, valid strictly only for a collision-dominated sheath [23]; and embedding the unresolved sheath as an effective electrical conductivity in the boundary-adjacent finite-volume cells of an otherwise quasi-neutral CFD solution, $\sigma_{\mathrm{eff}} = [(1 - I_g)/\sigma_{\mathrm{child}} + I_g/\sigma]^{-1}$ with σ_{child} a Child-law space-charge-limited conductivity [29]. We use a fourth strategy, implemented in the PLASMEX code: solve the same drift-diffusion– Poisson system, equations (1)–(13), in the bulk plasma and through the sheath alike, and use block-structured adaptive mesh refinement (AMR), built on the AMReX framework, to refine the mesh at the electrode down to a fraction of the local Debye length. No separate sheath closure, matching condition, or homogenized conductivity is introduced; the degree of local non-equilibrium is a property of the converged solution, not a modelling choice made in advance. This is closest in spirit to the unified modelling of [24], generalized from a specialized one-dimensional boundary-value solver to a general-purpose, AMR finite-volume PDE code, and it should be contrasted directly with the homogenized effective-conductivity shortcut of [29], which is the more practical choice for genuinely three-dimensional geometries where literal sub-Debye-length mesh refinement is not affordable.

4.2 Transport closures

Electron mobility and diffusion coefficients are obtained from a pre-tabulated, effectively Maxwellian two-term Boltzmann solution, $\mu_e N(T_e)$ and $D_e N(T_e)$. This is verified, not assumed: across the full tabulated range the ratio $(D_e/\mu_e)/T_e$ deviates from unity by 0.10% rms (worst case 0.50%), which is exactly the Einstein relation for a Maxwellian electron energy distribution, and confirms that at the 0.9–10% ionization fractions relevant here (ν_{ei}/ν_{en} ranging from 12 to 143 over $T_{\mathrm{gas}} = 1000$–$12\,000$ K) electron–electron and electron–ion collisions dominate every inelastic channel strongly enough to Maxwellize the bulk electron population, exactly as required by Section 3.8. The electron–ion Coulomb correction, equation (38), reduces the neutral-collision-only conductivity by one to three orders of magnitude and corrects its qualitative trend (neutral-only conductivity rises with temperature only because it divides by a falling neutral density; the corrected, Spitzer-like conductivity, $\propto T_e^{3/2}/\ln\Lambda$, falls in the $\sim 10^3$ S/m range expected for LTE argon at 1 atm). A Saha-equilibrium cross-check on the arc-core initial condition, $n_e^{\mathrm{Saha}} = 9.909 \times 10^{22}\,\mathrm{m}^{-3}$ against the solved $n_e = 9.860 \times 10^{22}\,\mathrm{m}^{-3}$, agrees to 0.5%, confirming the bulk solution is a genuine LTE state without that equilibrium being imposed. Ion transport uses the closed-form two-temperature Wannier closure of Section 2.1; at the reduced field found at the converged electrode cell of the sheath test case below (5976 Td), it gives $\mu/\mu_0 = 0.16$, a factor-of-six reduction from the low-field mobility, and recovers $\mu \to \mu_0$ in the low-field bulk.

4.3 Electrode boundary conditions

At the electrode face, the electron flux is set to the standard kinetic wall flux $\Gamma_e = \frac{1}{4} n_e \bar{c}_e$ evaluated on the (self-consistently depleted) wall density, and the ion flux uses a Hagelaar-type drift-aware closure, $\Gamma_i = \frac{1}{4} n_i \bar{c}_i + \max(\mu_i n_i E, 0)$, that reduces to the thermal flux at zero field and adds the outward drift contribution when the local field drives ions into the wall. This is structurally the same closure used for the counter-diffusing current in the NLTE-sheath boundary condition of [23] (their J_{CD}, equation A.7) and for the back-diffusion current in [29] (their J_{bd}, equation 11): a Boltzmann-retarded thermal flux evaluated at the wall density and local T_e. Two implementation choices in the present model depart from a fully symmetric, field-aware treatment of both species and are noted here as open items (Section 5 / `todo.txt`): the electron wall flux is currently evaluated with the field-independent $\frac{1}{4} n_e \bar{c}_e$ form rather than a Hagelaar-type field-aware form for electrons as well, and no secondary or thermionic electron emission term is included at the electrode, which is normally the dominant electron supply at a DC arc cathode and would need to be added before any cathode-fall prediction from this model could be considered complete.

4.4 Numerical implementation

The model is implemented in PlasmeX, a C++17 drift-diffusion plasma code built on the AMReX block-structured AMR framework, currently configured for a three-species (e^-, Ar^+, Ar) argon plasma. The electrostatic potential is solved with AMReX's geometric multigrid solver (MLMG) in general, and with Hypre's structured multigrid solver (`HYPRE_StructPFMG`) for the anisotropic two-dimensional axisymmetric mesh used in production runs, since MLMG's composite multigrid cycle was found not to converge on that mesh at any tested aspect ratio, bottom solver, or coarsening depth. Species transport uses a semi-implicit, operator-split scheme: drift fluxes are upwinded explicitly and diffusive fluxes are backward-Euler in an implicit face-coefficient form, $1 - \Delta t\,(e/\varepsilon_0)\mu_e n_e$, that removes the explicit dielectric-relaxation time-step restriction of a fully explicit electron drift treatment and gives roughly a 100× larger stable time step at equal accuracy, verified by matching both schemes at matched physical time to $\sim 10^{-6}$ relative agreement after a long integration from a common checkpoint.

In one spatial dimension, both linear solves (the electrostatic Poisson solve and the implicit species-diffusion solve) reduce to a tridiagonal system on each AMR level, for which a direct solver is both faster and more accurate than the general-purpose iterative multigrid solver: instrumented at 93% of a time step for a 512-cell problem, the iterative solver is replaced by a direct Thomas-algorithm solve whose matrix entries are recovered once, exactly, by probing the discrete operator, giving an 8.9× reduction in total time-step cost (65× and 13× on the potential and species solves individually) with an exact rather than iteratively converged linear-algebra answer. This extends, non-trivially, to the full composite AMR hierarchy: refluxing at a coarse-fine interface exactly cancels the covered coarse cell and replaces it with the fine-level flux, and the coarse-to-fine ghost-cell interpolation used in one dimension is an exact two-term closure over a single coarse cell, ghost $= \frac{2}{3}x_C + \frac{1}{3}x_{f_0}$ at a refinement ratio of 2 — together these guarantee that the composite operator over the leaf cells of a refined 1D hierarchy is itself exactly tridiagonal, so the same direct solve applies unmodified across level boundaries. This direct solver is what makes literal AMR refinement to sub-Debye-length cells in the sheath computationally practical rather than prohibitively expensive; it is restricted to one spatial dimension and a single MPI rank, so the production two-dimensional axisymmetric configuration falls back to Hypre as noted above.

5 Results

6 Conclusion

Acknowledgments

To be completed.

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